

Jayden Patel

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EDUCATION

The University of Texas at Austin Bachelor of Science, Mechanical Engineering August 2023 – May 2027
Minor: Robotics (admitted in a small cohort)
Overall GPA: 3.92

Relevant Coursework: Heat Transfer, Mechanics of Solids, Materials Engineering, Mechatronics, Fluid Mechanics, Systems and Controls, Dynamics, Thermodynamics, Statics, Programming and Eng. Comp. Methods (MATLAB)

EXPERIENCE

Boeing (Defense, Space & Security) – *Space Electronics Product Design Intern*; El Segundo, CA May 2026 – August 2026

- Designed a **12-layer Megtron 7GN PWB** backplane for satellite flight hardware, integrating electrical routing with mechanical packaging and hardware interface constraints in Siemens Xpedition Layout and Creo
- Optimized board layout, via structures, and layer transitions to eliminate backdrilling and reduce fabrication costs by 20%
- Defined **mechanical keepouts, contact regions, and board-level constraints** while performing DFM checks to ensure reliable integration of the PWB within the flight hardware assembly

Boeing (Defense, Space & Security) – *Mechanical Product Design Intern*; El Segundo, CA June 2025 – August 2025

- Engineered and structurally validated an **8-ft payload lifting system** in Creo and ANSYS for satellite test operations, ensuring safe handling and positioning of **80-lb flight hardware** under strict motion and operator safety constraints
- Achieved 6-ft vertical travel, 1-inch lateral positioning tolerances, and 10° rotational alignment for precision payload placement
- Conducted **screw length analysis, bolt calculations, and tolerance stack-ups** to ensure mechanical integrity and reliable assembly
- Supported vibration testing for electronic satellite products, ensuring component response stays consistent during simulated forces

UT Engines Research Group – *Research Assistant*; Austin, TX January 2025 – December 2025

- Research focused on **quantifying electrode erosion for different metallic alloys** in spark plugs under different operating conditions
- Performed in spark calorimeter, measuring breakdown voltage, arc motion, and current fluctuations under pressures up to 300 psi
- Developed ignition models to optimize spark plug performance and efficiency leading to more fuel-efficient engines

Credo Semiconductor, Inc. – *Mechanical Engineering Intern*; San Jose, CA May 2024 – August 2024

- Partnered with Internal Applications COB team to design and manufacture new QSFP-DD shells to eliminate third party vendors
- Created **thermally stable mechanical fixtures for multi-PCB regression setups** and achieved solution 25% under budget

LEADERSHIP EXPERIENCE AND ACTIVITIES

Longhorn Baja Racing SAE – *Dynamics Engineer, Simulation & Validation Engineer* January 2026 – Present

- Developed **brake rotor and wheel hub assemblies for an off-road vehicle**, including material selection and manufacturing planning
- Reduced brake rotor mass by 30% through geometry optimization while **maintaining structural and transient thermal performance**
- Evaluated laser cutting and machine processes, **analyzing heat-affected zones to prevent premature failure** in load-bearing regions
- Validated designs through manufacturability and structural reports, preventing costly rework within a \$25,000 vehicle budget

Texas Guadalupe – *Dynamics Engineer* January 2025 – August 2025

- Designed and manufactured hyperloop bogie incorporating levitation technology to optimize stability and performance
- Architected a **fail-safe pneumatic braking system** using spring-applied calipers to ensure pod deceleration in emergency scenarios
- 2x SpaceX Hyperloop Pod Competition - Innovation Award winner, National Instruments Engineering Impact Award winner

UT IEEE Robotics and Automation Society – *Mechanical Engineering Lead* February 2024 – May 2025

- Led a team of 5 engineers to develop and construct fleet of robots to compete in different battle scenarios; Robomaster game
- Optimized the Sentry robot turret architecture into **two independent 360° rotating turrets**, increasing projectile capacity by 29% while achieving full-field defensive coverage
- Significant use of CAD through SolidWorks and hardware manufacturing, ensuring functional designs were operational

ADDITIONAL INFORMATION

Computer Skills: Creo, Siemens NX, Mentor Siemens Xpedition, Certified SolidWorks Designer, ANSYS, MATLAB, Finite Element Analysis (FEA), Machine Shop Manufacturing, Proficient with Microsoft Office Tools

Involvement: Tau Beta Pi Engineering Honor Society, The American Society of Mechanical Engineers (ASME), Faith Café (Volunteering), Desi Students of Texas (DST), Beta Theta Pi Fraternity

Certifications & Honors: SolidWorks Associate Mechanical Design Certification (2025), Dean's Honors List (6 Semesters), International Baccalaureate (IB) Diploma (May 2023), AP Scholar with Distinction (2022, 2023)